RIPHAH INTERNATIONAL UNIVERSITY ISLAMABAD

Faculty of Social Sciences & Humanities

Department of Computer Science

Final Term Examination Spring 2026 Semester

Course Name:	Object-Oriented Programming	**Course Code:**	CS201
Program:	BS (Computer Science)	**Section:**	C
Semester:	Final	**Date of Examination**	01-06-2026
Teacher's Name	Muhammad Usman	**Total Marks**	40
Exam Duration	(10:00 am – 12:00 pm)		

INSTRUCTIONS

- Write your name and SAP ID on all answer sheets.
- Write page number on all sheets.
- Please be analytical and specific in answering questions.
- Switch off your Mobile Phone.
- Attempt all the following questions critically and creatively.

Question 1: **[12 Marks]**

Answer the following short scenario-based questions:

(a) A programmer writes a single class with 2000 lines performing all operations. Which OOP principles is he violating and why? How would you restructure this design? [3 Marks]

(b) Explain constructor chaining with an example. What problem does it solve in object initialisation? [3 Marks]

(c) What is the role of the 'final' keyword in Java? How does it relate to inheritance and polymorphism? Use examples to support your answer. [3 Marks]

(d) Differentiate between aggregation and composition with real-world examples. Why does this distinction matter in system design? [3 Marks]

Question 2: **[08 Marks]**

(a) Identify and correct ALL errors in the following code:

```
class BankAccount {
    private double balance;

    public BankAccount(double initialBalance) {
        balance = initialBalance;
    }

    public void withdraw(double amount) {
        if (amount > balance)
            System.out.println("Insufficient funds");
        else
            balance -= amount;
```

```java
            System.out.println("Withdrawn: " + amount);
    }

    public double getBalance { return balance; }
}

class Main {
    public static void main(String[] args) {
        BankAccount acc = BankAccount(500);
        acc.withdraw(100);
        System.out.println("Balance: " + acc.getBalance());
    }
}
```

(b) Analyze the following code and write the exact output:

```java
class Shape {
    String color;
    Shape(String c) { this.color = c; }
    double area() { return 0; }
}

class Circle extends Shape {
    double radius;
    Circle(String c, double r) { super(c); radius = r; }
    double area() { return 3.14 * radius * radius; }
}

class Main {
    public static void main(String[] args) {
        Shape s1 = new Circle("Red", 5);
        Shape s2 = new Shape("Blue");
        System.out.println(s1.color + ": " + s1.area());
        System.out.println(s2.color + ": " + s2.area());
    }
}
```

Question 3: **[20 Marks]**

Design a University Course Registration System in Java incorporating the following OOP concepts:

1. Inheritance: Create a base class User and derive Student and Instructor from it.
2. Encapsulation: Protect sensitive data like GPA and contact details using private access with getters/setters.
3. Polymorphism: Override a generateReport() method for Student and Instructor to display relevant info.
4. Abstraction: Use an abstract class or interface SystemUser defining common behaviours.
5. File Handling: Save registered course records to a file and retrieve them on program startup.
6. Exception Handling: Handle duplicate registration, file errors, and invalid CGPA values using try-catch.

Note: The system should allow a student to register for courses, view registered courses (loaded from file), and display a report.